DATABASE DESIGN AND DEVELOPMENT (IT2140)

LECTURE 03 – SQL JOINS

LECTURE CONTENT

- Introduction to Joins
- Inner Join
- Outer Join (Left/Right)

LECTURE CONTENT

- At the end of this lecture students should be able to
  - Write syntactically correct SQL join statements to combine and retrieve data from multiple tables in an RDBMS.

[INNER] JOIN

- Joins are used to combine data from two or more tables based on a related column (usually Primary Key–Foreign Key relationship).
- Ex : Find names of students and their program offerings.

```
Select s.Name, p.OfferBy
From Student s, Program p
where s.pid=p.progId
        Or
SELECT s.Name, p.OfferBy
FROM Student s INNER JOIN Program p
      ON s.pid = p.progID;
```

Student

SID	Name	gpa	pid
1000	Amal	3.2	BM
1001	Nimali	2.8	IT
1002	Aruni	3.8	SE
1003	Surani	2.5	IT

Program

progId	years	Offer By
BM	3	Curtin
IT	4	SLIIT
SE	3	SHU

Name	offerBy
Amal	Curtin
Nimali	SLIIT
Aruni	SHU
Surani	SLIIT

[INNER] JOIN WITH CONDITIONS

- Joins two tables based on a certain condition
- Ex : Find the names of students who follow programs offered by SLIIT

```
Select s.Name
From student s, program p
where s.pid=p.ProgId and offerBy='SLIIT'
```

Or

```
Select s.Name
From student s INNER JOIN program p on
    s.pid=p.progId
Where offerBy='SLIIT'
```

Student

SID	Name	gpa	pid
1000	Amal	3.2	BM
1001	Nimali	2.8	IT
1002	Aruni	3.8	SE
1003	Surani	2.5	IT

Program

progId	years	Offer By
BM	3	Curtin
IT	4	SLIIT
SE	3	SHU

Name
Nimali
Surani

LEFT OUTER JOIN

- Returns all rows from the table on the left-hand side of join, with the matching rows in the table on the right-hand side of the join.
- The result is NULL in the right side when there is no match.
  - Ex : For all the students display the name and the offering institute

Select s.Name, p.offerBy

From student s LEFT OUTER JOIN program p

on s.pid=p.progId

Student

SID	Name	gpa	pid
1000	Amal	3.2	BM
1001	Nimali	2.8	IT
1002	Aruni	3.8	SE
1003	Surani	2.5	IT

Program

progId	years	Offer By
BM	3	Curtin
IT	4	SLIIT

Name	offerBy
Amal	Curtin
Nimali	SLIIT
Aruni	NULL
Surani	SLIIT

RIGHT OUTER JOIN

- Returns all rows from the table on the right-hand side of join, with the matching rows in the table on the left-hand side of the join.
- The result is NULL in the left side when there is no match.
  - Ex : For all the programs display the offering institute and names of the students following

Select s.Name, p.offerBy

From student s RIGHT OUTER JOIN program p on s.pid=p.progId

Student

SID	Name	gpa	pid
1000	Amal	3.2	BM
1001	Nimali	2.8	IT
1002	Aruni	3.8	NULL
1003	Surani	2.5	IT

Program

progId	years	Offer By
BM	3	Curtin
IT	4	SLIIT
SE	3	SHU

Name	offerBy
Amal	Curtin
Nimali	SLIIT
Surani	SLIIT
NULL	SHU

WHAT YOU HAVE TO DO BY NEXT WEEK

- Try out the self-test questions on the courseweb.
- Complete the tutorial.